

1.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main()
{
    char command[100];

    printf("Enter Linux command: ");
    fgets(command, sizeof(command), stdin);

    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0)
    {
        perror("Fork failed");
        exit(1);
    }
    else if (pid == 0)
    { printf("\nChild Process\n");
      printf("Child PID: %d\n", getpid());
      printf("Parent PID: %d\n", getppid());

      execlp(command, command, NULL);

      perror("Execution failed");
      exit(1);
    }
    else
    {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
    }
}
```

```

        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}

```

```

Child Process
Child PID: 2201
Parent PID: 2200
Parent Process
Parent PID: 2200
Child PID: 2201
MyProject      exec      filecopy.c  os.c      os1.2.c  os1.c    os14.c  os2.c   os4.c   os6.c   os8.c      test.txt
destination.txt filecopy  os       os1.1.c  os1.3.c  os1.xc   os15.c  os3.c   os5.c   os7.c   source.txt
Child process completed.

```